

REVISED RESULTS

VC Scout

Revised Results, Insights Summary, and Implementation Plan

Startup Growth and Investment | Expanded Dataset Audit + Audited Benchmark Model

75,230

expanded rows audited

950

unit-suspect funding rows
flagged/corrected

1,829

rows with valuation

0.30

audited test R^2

0.49

funding elasticity

77%

noisy Form D mega dollars

Executive answer

VC Scout should be implemented as a benchmark-and-scouting framework, not a startup-success prediction engine. The revised work audits the expanded data, preserves traceability, and shifts the insight from raw valuation to funding-adjusted outperformance.

Team 17

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Submission package includes PPTX/PDF plus auditable CSVs, source-of-truth JSON, and a reproducible Python script.

What changed since the prior submission

The prior work established EDA and a first valuation model; this submission turns that into a defensible implementation plan.

Refined tools and models

We landed on a two-layer architecture: an audited unicorn valuation benchmark plus an expanded-tier diagnostic layer. Gradient Boosting remains the benchmark champion, while OLS/Ridge remain interpretability checks.

Shortcomings addressed

The expanded data adds useful controls, but it also introduces funding-unit risk, source/timing mismatch, missing valuation coverage, and noisy Form D entities. We address these through flags, corrections, and decision boundaries.

Insights and hypotheses

The analysis now tests 2021 as a market-regime shock, validates diminishing returns to funding, stress-tests no-funding scouting, and uses residuals to identify outperformance patterns.

Business implications

VC Scout should rank segments by benchmark outperformance, not raw valuation or raw unicorn count. This reduces hype bias and makes rankings more explainable.

Implementation plan

The final deliverable should include a transparent scoring layer, sample-size confidence flags, investor network features, and a dashboard/dashboard-style slide for business users.

Traceability standard

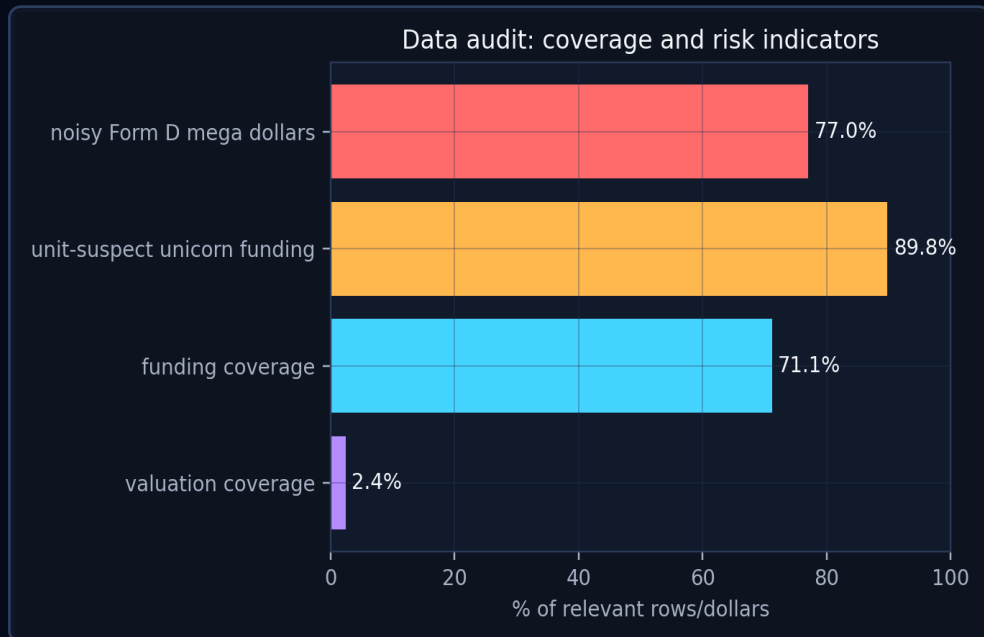
Every number in the deck maps to a cleaned CSV, model output file, source-of-truth JSON, or reproducible script. The goal is to avoid black-box reporting.

Framing shift

From predict unicorn valuation to identifying funding-adjusted valuation outperformance signals among unicorn and comparable startup tiers.

Expanded data improves the project only after an audit

This is where the revised analysis directly addresses data quality and survivorship-bias risk.



Coverage limitation

Only 1,829 of 75,230 rows have valuation. This prevents valuation modeling across the full startup universe.

Funding unit risk

950 unicorn rows were flagged because funding exceeded valuation, indicating a likely M/B scaling issue in the expanded build.

Form D noise

Raw Form D includes many funds, real estate entities, and finance vehicles. It is retained as market context, not a clean startup-training set.

Survivorship remains

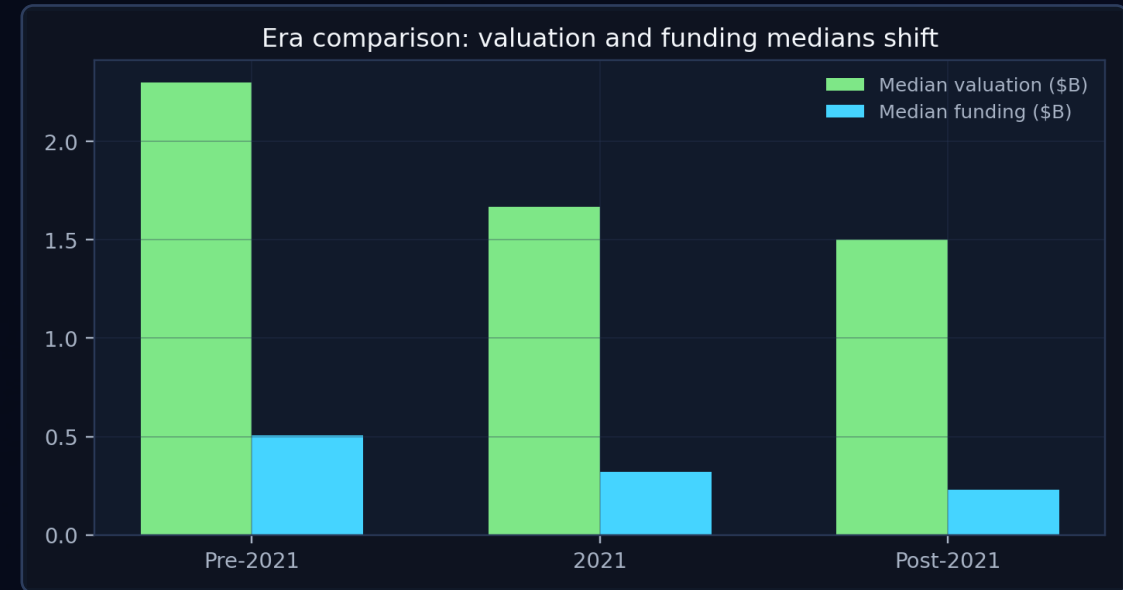
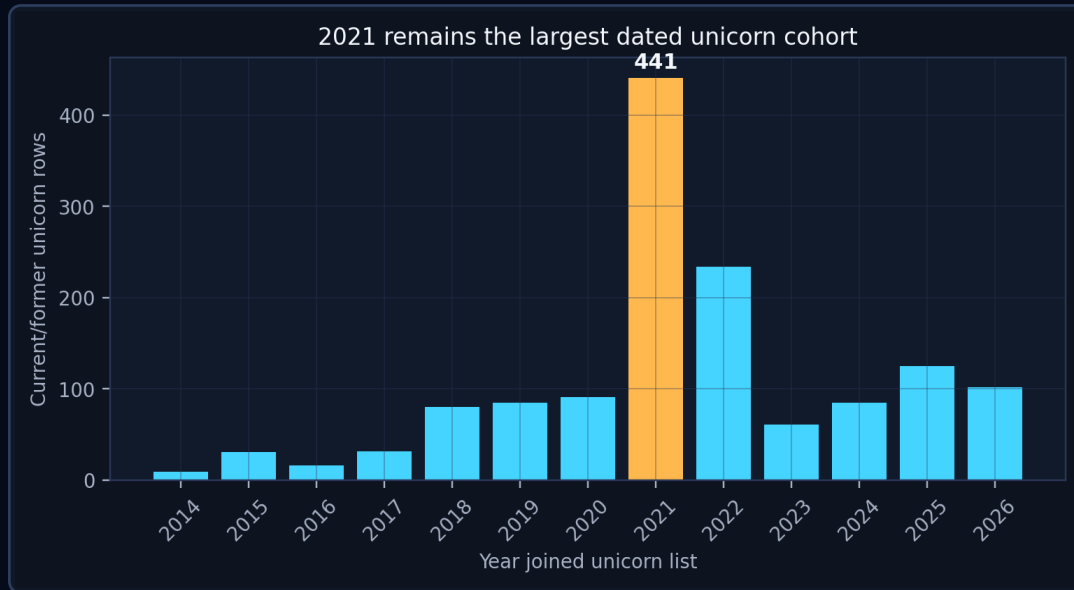
The expanded master adds comparison tiers but is not a time-aligned panel of all startups from formation to outcome.

How we addressed it

The cleaned master preserves raw values, adds audited funding, assigns audit flags, isolates Form D, and separates unicorn valuation benchmarking from broader tier diagnostics.

2021 is a market-regime shock, not a normal trend line

The data flags the discontinuity; it does not prove COVID causality by itself.



Evidence line: Kruskal-Wallis $p < 0.001$ for valuation and funding by era; χ^2 $p < 0.001$ for sector mix; time-to-unicorn is not significant ($p = 0.267$).

What is proven

2021 is the largest single dated current-unicorn cohort in the expanded data, and era differences in valuation, funding, and sector mix are statistically significant.

What is not proven

The dataset alone cannot separate COVID-driven demand, cheap capital, investor behavior, or valuation inflation as causal explanations.

Action for VC Scout

Keep a 2021/era indicator and show users when a benchmark is influenced by an unusual funding regime.

Two-layer analytics architecture

The tools are chosen to match what the data can support, not what sounds most sophisticated.

Layer A: audited unicorn valuation benchmark

Supervised regression on $\ln(\text{valuation})$ for unicorn-history rows with audited funding. Gradient Boosting captures nonlinearity; OLS/Ridge provide explainability and elasticity checks.

Layer B: expanded-tier diagnostic model

Classifier compares unicorn-history companies with funded controls, accelerator controls, and high-funding proxies. It is useful for screening patterns, not causal winner prediction.

Benchmark

Residuals

Scoring

Decision

Model output

Expected valuation given funding, sector, geography, era, tier, speed, and investors.

Insight output

Actual minus expected valuation identifies funding-adjusted outperformance.

VC Scout score

Blend residuals, tier signal, sample size, data confidence, and investor network features.

User action

Prioritize sector/region/investor clusters for deeper diligence, not automatic investing.

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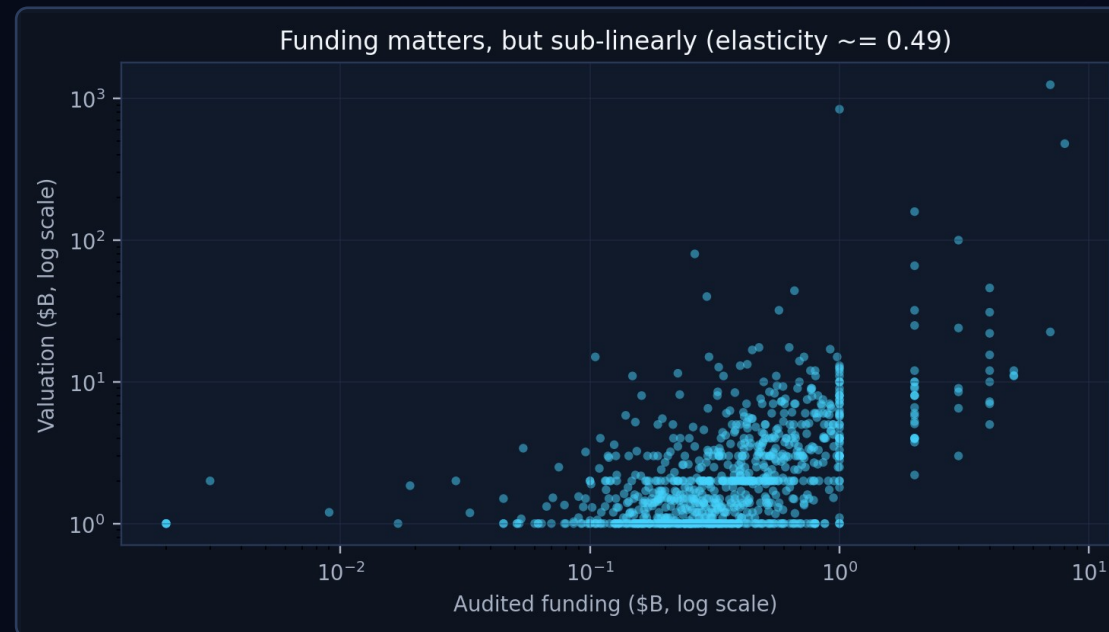
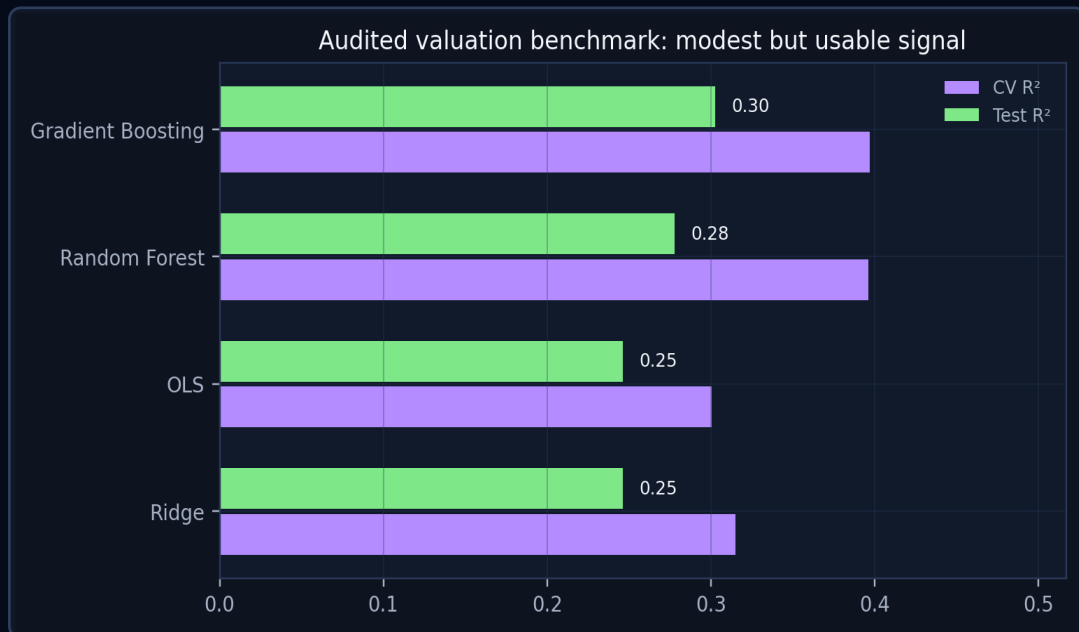
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Guardrail

No claim of "probability of becoming a unicorn" until there is a true time-aligned non-unicorn panel with observed outcomes.

The audited benchmark has modest but credible signal

The important result is not a perfect model; it is a defensible benchmark engine.



Champion

Gradient Boosting remains preferred after the audit: test $R^2 \approx 0.30$. This is lower than the earlier Kaggle-only result, but more defensible.

Funding effect

Funding remains important, but elasticity ≈ 0.49 means valuation rises less than one-for-one with capital raised.

No-funding stress test

Removing funding drops R^2 to about 0.05, showing early-stage scouting is much harder than late-stage benchmarking.

Implication

Use model residuals and segment signals, not exact point estimates.

Hypotheses proven, disproven, and carried forward

This turns the analysis into decision-ready learning, not just charts.

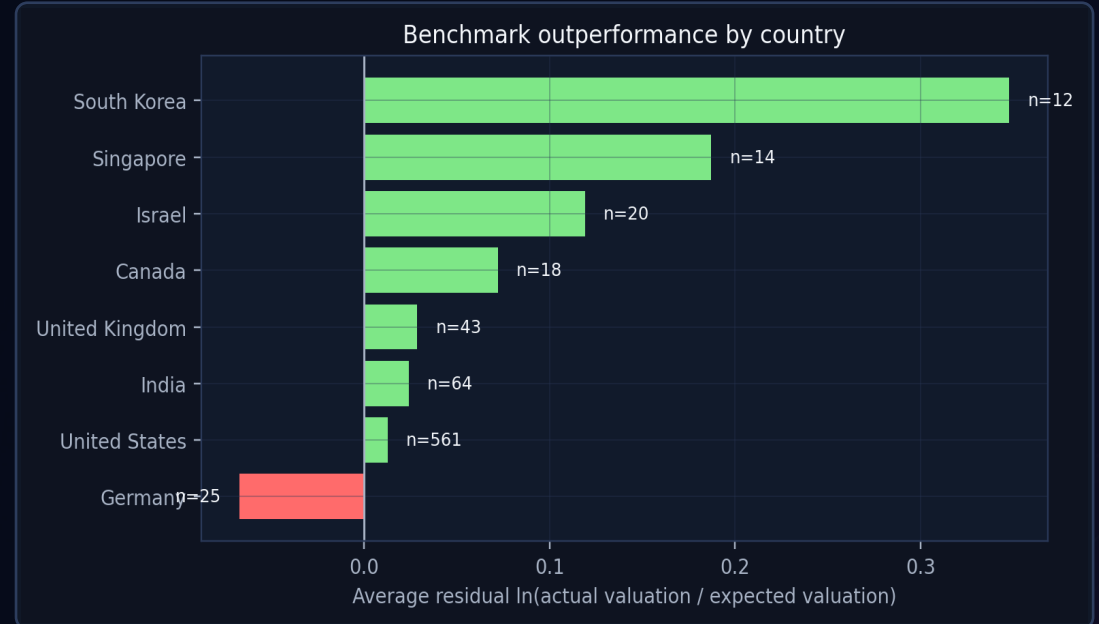
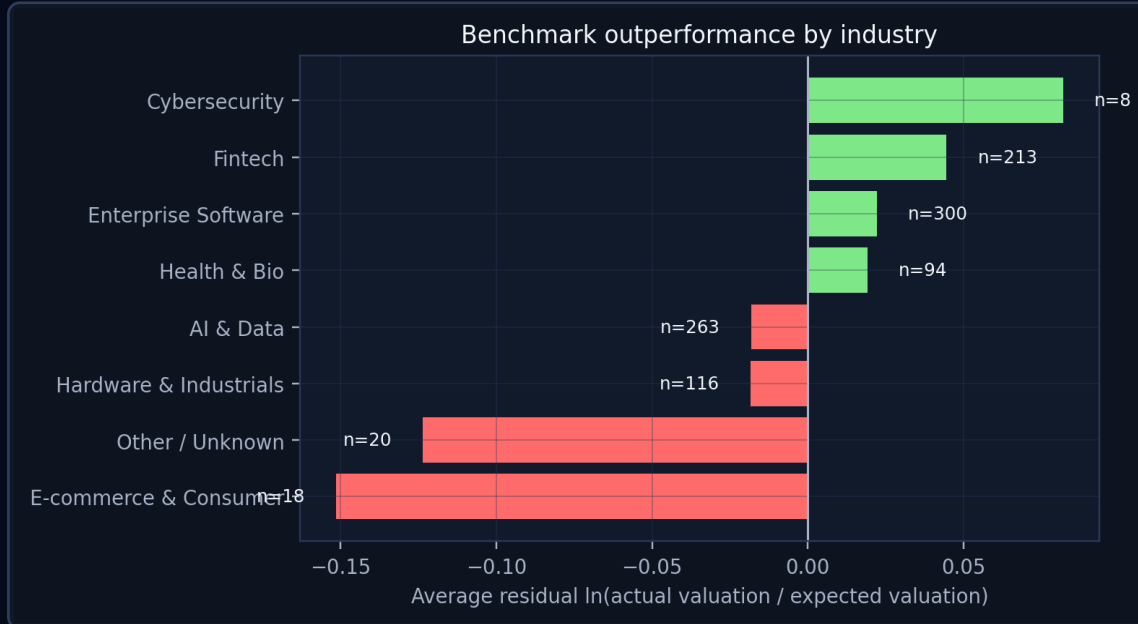
Hypothesis	Status	Evidence / implication
H1: 2021 is a distinct regime	Supported	Largest dated cohort; valuation/funding distributions and sector mix differ by era. Interpret as market-regime signal, not proof of COVID causality.
H2: Funding has diminishing returns	Supported	Audited log-log elasticity ≈ 0.49 . More capital helps, but raw funding does not scale valuation one-for-one.
H3: Sector/geography add signal	Partially supported	Residual tables show directional variation, but thin samples need confidence flags before ranking.
H4: Expanded controls solve survivorship bias	Disproven / partial	The 75k-row master helps context, but source timing, missing valuation, and proxy labels remain.
H5: Investor count captures network quality	Not proven	Investor count is too crude; final model should use top-fund flags and co-investor clusters.

Next analytic emphasis

Move from company-level valuation prediction to segment-level outperformance, where data quality and sample-size caveats can be surfaced honestly.

Residuals turn a model into VC Scout

Actual minus expected valuation is the cleanest bridge from statistics to business action.



How to read it

Positive residuals mean a segment exceeds expected valuation after adjusting for funding and context.

What is useful

Residuals create watchlists: segments worth deeper diligence because they outperform a fair benchmark.

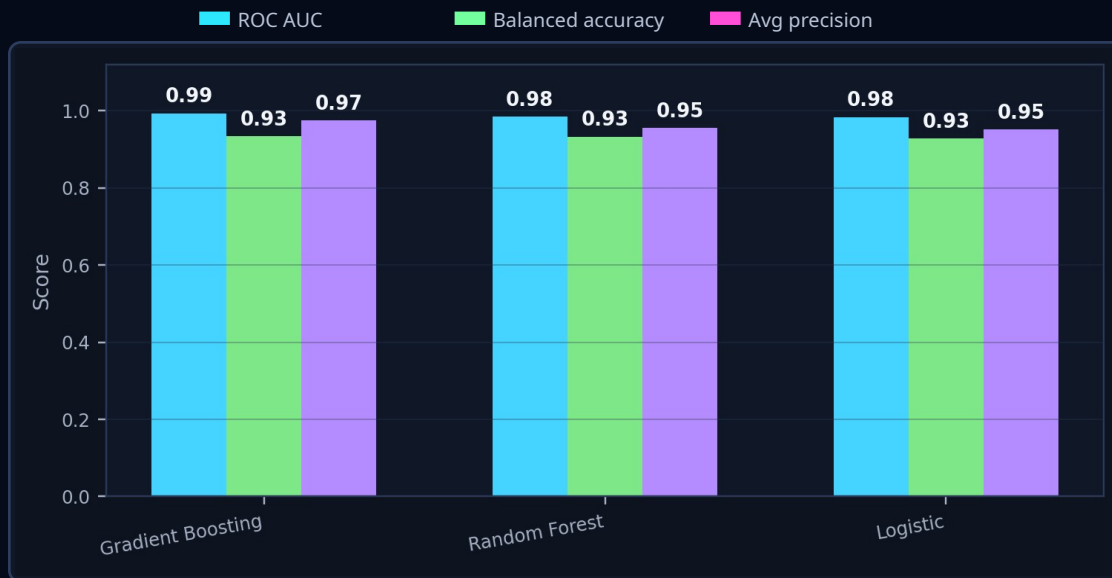
What to avoid

Do not treat small-n residual leaders as final rankings. Add confidence and sample-size penalties.

High accuracy can be a warning sign

The classifier can screen patterns, but it can also learn source artifacts.

Expanded-tier classifier: high diagnostic accuracy, high bias risk



Tempting read

The model separates unicorn-history rows from controls with very high diagnostic accuracy.

Statistician read

Some of that signal is likely source/timing bias, not true startup destiny.

Safe use

Use expanded-tier scores to prioritize sectors and questions for diligence.

Unsafe use

Do not claim this predicts arbitrary startups becoming unicorns.

Implementation decision

The expanded dataset belongs in VC Scout as a diagnostic comparison layer and confidence input, not as the sole basis for a success-probability model.

Initial recommendations for a VC Scout user

The recommendations are deliberately decision-support oriented.

1. Rank outperformance, not size

Raw valuation and raw unicorn count are too sensitive to capital intensity and market cycles. Rank segments against expected valuation benchmarks.

2. Keep 2021 visible

Treat 2021 as a market-regime flag. Do not let one unusual funding period define “normal” scouting baselines.

3. Use residuals as watchlists

Sectors, countries, or investor clusters with repeated positive residuals become candidates for deeper diligence.

4. Add confidence labels

Every score should show sample size, data coverage, and whether the signal depends heavily on audited/corrected fields.

5. Upgrade investor features

Replace investor count with top-fund flags, sector specialization, co-investor networks, and timing of involvement.

6. Separate model from diligence

The tool narrows the search space. Final investment judgments still require revenue, retention, burn, TAM, and ownership data.

Recommended positioning

VC Scout is a market-intelligence and prioritization tool, not an automated investment recommender.

What gets built before in-person week

The final submission should be traceable enough that every number can be defended.

Sprint	Workstream	Concrete output
1	Data QA hardening	Finalize audited master, funding audit rule, source notes, and Form D exclusion/context logic.
2	Benchmark model	Lock audited valuation benchmark, residual outputs, and no-funding stress test.
3	VC Scout score	Combine residual outperformance, tier signal, sector/geography medians, sample-size penalty, and data confidence.
4	Dashboard / slideware	Build final user-facing view: sector, region, investor cluster, confidence, and limitations.
5	Final review	Package PPTX/PDF, cleaned CSVs, source-of-truth JSON, and reproducible script for auditability.

Deliverable structure

PowerPoint/PDF for stakeholders; audit-trail ZIP for the professor or anyone asking for raw analysis.

Final model language

“Funding-adjusted valuation outperformance signal” stays defensible and avoids unsupported causal success claims.

Remaining data need

A true success model requires a time-aligned startup panel with non-unicorns, failures, acquisitions, and follow-on outcomes.

Audit-trail files

Audited master CSV, Form D audited CSV, model results CSV, residual tables, hypothesis summary, source-of-truth JSON, and analysis script.